

# Value Chain Physics: Governance Laws of Open Complex Giant Systems

— A Complete Physics Constitution, Formal Operators, and 100-Billion-Scale Industrial Validation

**Fanchun Meng (Grit Meng)**

General Design Department of Global Digital Neural System Centroid

Former Chief Architect of Global Supply Chain Planning Systems (IPS), Lenovo Group

Email: gritmeng@outlook.com | GitHub: <https://gritmeng.github.io/Value-Chain-Physics/>

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## Abstract

Modern large-scale discrete manufacturing networks and global value chains represent classic Open Complex Giant Systems (OCGS), characterized by non-independent and identically distributed (Non-IID) interactions and factorial state-space explosions ( $O(N!)$ ). Traditional management paradigms—relying on steady-state assumptions, local optimization, and open-loop administrative workflows—inevitably collapse under Wolfram computational irreducibility and organizational entropic dissipation. In this monograph, we establish **Value Chain Physics (VCP)**, an axiomatic theoretical framework for open complex giant systems. Reconstructing Qian Xuesen’s “metasynthesis” through first-principles physics, we define the Digital Double-Helix Ontology  $\langle \mathcal{D}, A \rangle$ , coupling a five-dimensional orthogonal value-chain container  $\mathcal{D}$  with a discrete-time algorithm engine  $A$ . We prove five foundational theorems: (1) algebraic irreducibility and topological completeness of container  $\mathcal{D}$ ; (2) the quadratic decay law of Return on Invested Capital (ROIC) with respect to manufacturing lead time; (3) non-linear WIP inflation under non-kit arrival delays; (4) absolute optimality of physical-constraint truncated IF-THEN-ELSE stepping search; and (5) governance as a necessary mathematical precondition for observability. For engineering execution, we deploy Data-Oriented Design (DOD) with SIMD bitset vectorization in the IPC engine, achieving a decisive empirical benchmark: resolving 500,000 discrete demand orders, 2,000,000 SKU-location nodes, and 150,000 physical constraints in 296 seconds (5 minutes) on a single commodity personal computer (PC) for LCFC (World Economic Forum Lighthouse Factory). Finally, we demonstrate a 1:1 mathematical isomorphism between VCP and five classical system theories (Prigogine Dissipative Structures, Ashby Requisite Variety, Karl Friston Free Energy Principle, Conant-Ashby Good Regulator Theorem, and Renormalization Group Dynamics), elevating value chain governance from heuristic management to an empirical, falsifiable hard physics science.

**Keywords:** Value Chain Physics; Open Complex Giant Systems; Digital Double-Helix On-

tology; Computational Irreducibility; ROIC Quadratic Decay; Kalman Observability; Human-Out-of-the-Loop; Free Energy Principle

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# 1 Introduction: Mechanical and Information Limits under Algorithmic Gravity

The management paradigms of human industrial civilization have experienced two major revolutions, and are now entering a third era driven by high-dimensional complexity absorption:

1. **The Mechanical Era (Marked by Frederick Taylor’s ”Scientific Management”)**: Focused on standardizing physical labor, solving ”how to make human physical actions as efficient as machines,” propelling productivity from craft workshops to industrial assembly lines.
2. **The Information Era (Marked by Peter Drucker’s ”Modern Management” and ERP Systems)**: Focused on bureaucratic synergy and knowledge work motivation, solving ”how to make human brains work for common commercial goals,” giving rise to twentieth-century multinational corporations.

However, as global manufacturing networks deepen, modern production networks have evolved into **Open Complex Giant Systems (OCGS)** comprising hundreds of plants, millions of SKUs, dynamic rush orders, component substitutions, and shared line capacities. In this high-dimensional gravity field, traditional paradigms experience systemic physical collapse:

- **Organizational Entropy & Communication Friction**: Human concurrent processing bandwidth is bounded by Miller’s limit ( $7 \pm 2$ ). Across multi-departmental networks, communication complexity explodes as  $O(K^2)$ . Without unified physical constraints, departments lock into local Nash equilibria, generating massive organizational ”friction heat” while net forward work approaches zero.
- **Goodhart Collapse of Probabilistic AI**: Black-box AI models lacking physical priors suffer from Goodhart’s Law under non-stationary distributions. Optimizing low-dimensional reward metrics leads to proxy arbitrage and hallucinations, causing catastrophic real-world delivery failures.
- **ERP/APS Two-Layer Disconnection**: Traditional ERP/APS systems rely on steady-state, I.I.D., and unconstrained capacity assumptions. Facing multi-level BOM entanglements, their computation overflows to  $O(N!)$ , leading to complete decoupling between digital plans and shop-floor execution.

In 1990, Prof. Qian Xuesen introduced ”Open Complex Giant Systems” and stated that the only path forward is ”combining qualitative and quantitative metasyntesis through human-machine integration.” Building on Qian’s vision and 22 years of field execution across Lenovo’s global manufacturing network, we establish **Value Chain Physics (VCP)**, unifying thermodynamics, information theory, cybernetics, and bare-metal algorithms into a rigorous physics constitution for enterprise governance.

## 2 Physical and Mathematical Foundation of Value Chain Physics

### 2.1 The 5D Value Chain Container $\mathcal{D}$

We project the spatiotemporal state of value chain networks onto a five-dimensional orthogonal topological manifold:

$$\mathcal{D} = \langle V_D, E_D, C_D, T_D, \mathbf{x}_D \rangle \quad (1)$$

where:

1. **Node Set  $V_D$ :**  $V_D = \{v_1, v_2, \dots, v_n\}$  representing physical, contractual, and financial mass points (plant capacities, warehouse bins, supplier nodes).
2. **Directed Topology  $E_D$ :**  $E_D \subset V_D \times V_D \times \mathcal{M}_{\text{BOM}}$  representing material, contractual, and financial propagation trajectories over multi-echelon Bill of Materials (BOM) matrices.
3. **Constraint Cluster  $C_D$ :**  $C_D = \{g_1(\mathbf{x}_D) \leq 0, g_2(\mathbf{x}_D) = 0, \dots\}$  defining physical capacity limits, lead-time shift damping, material conservation equations, and channel allotments.
4. **State Transition Operator  $T_D$ :**  $T_D : \mathcal{X}_D \times \mathcal{E}_{\text{event}} \longrightarrow \mathcal{X}_D$  governing discrete state jumps triggered by real-world events.
5. **State Vector  $\mathbf{x}_D$ :**  $\mathbf{x}_D(t) = [I(t), D_{\text{gra}}(t), P_{\text{eg}}(t)]^T$  capturing inventory snapshots, demand gravity waves, and causal pegging matrices.

### 2.2 Theorem 1: Algebraic Irreducibility and Topological Completeness

**Theorem 2.1.** *The five-dimensional value chain container  $\mathcal{D} = \langle V_D, E_D, C_D, T_D, \mathbf{x}_D \rangle$  is algebraically irreducible and topologically complete for governing open complex giant systems.*

*Proof. 1. Topological Completeness:* The state trajectory of any open complex manufacturing system is defined by structure  $\mathcal{G}_{\text{struct}}$ , boundary conditions  $\mathcal{B}_{\text{bound}}$ , and dynamics  $\mathcal{F}_{\text{dyn}}$ . In  $\mathcal{D}$ ,  $\mathcal{G}_{\text{struct}} \equiv (V_D, E_D)$ ,  $\mathcal{B}_{\text{bound}} \equiv C_D$ , and  $\mathcal{F}_{\text{dyn}} \equiv (T_D, \mathbf{x}_D(t))$ . Because  $\mathcal{D}$  forms a Mutually Exclusive, Collectively Exhaustive (MECE) decomposition of state dynamics, no physical trajectory falls outside  $\mathcal{D}$ .

**2. Algebraic Irreducibility:** Removing any single dimension  $d_k \in \{V_D, E_D, C_D, T_D, \mathbf{x}_D\}$  causes a catastrophic loss of control or observability:

- Removing  $V_D$  or  $E_D$  collapses the transfer function  $H(s) \rightarrow 0$ , destroying controllability.
- Removing  $C_D$  expands the feasibility boundary  $C_D \rightarrow \emptyset$ , producing physically unexecutable control vectors  $\mathbf{u}(t)$ .
- Removing  $T_D$  reduces dynamics to  $\frac{d\mathbf{x}_D}{dt} = 0$ , rendering the system incapable of dynamic adaptation.

- Removing  $\mathbf{x}_D$  reduces the Kalman observability Gramian rank to zero ( $\text{rank}(\mathcal{O}) = 0$ ), opening the control loop.

Hence, container  $\mathcal{D}$  is minimal and irreducible.  $\square$

### 2.3 Theorem 2: Quadratic Decay Law of ROIC

**Theorem 2.2.** *In To-Order manufacturing (MTO/ATO/CTO), Return on Invested Capital (ROIC) decays quadratically with respect to shop floor manufacturing lead time  $LT_{\text{shop}}$ :*

$$\frac{dROIC}{dLT_{\text{shop}}} = -ROIC^2 \cdot \left( \frac{COGS}{365 \cdot NOPAT} \right) \quad (2)$$

*Proof.* By financial physics,  $ROIC = \frac{NOPAT}{IC}$ , where  $IC = I_{\text{WIP}} + I_{\text{RM}} + AR - AP + FA$ . By Little's Law, work-in-process inventory is  $I_{\text{WIP}} = COGS \cdot \frac{LT_{\text{shop}}}{365}$ . Differentiating ROIC with respect to  $LT_{\text{shop}}$ :

$$\frac{dROIC}{dLT_{\text{shop}}} = -\frac{NOPAT}{IC^2} \cdot \frac{dIC}{dLT_{\text{shop}}} = -\frac{NOPAT}{IC^2} \cdot \frac{COGS}{365} \quad (3)$$

Substituting  $IC = \frac{NOPAT}{ROIC}$  yields the exact quadratic decay differential equation.  $\square$

### 2.4 Theorem 3: Non-Linear WIP Inflation under Non-Kit Wait Times

**Theorem 2.3.** *In multi-echelon BOM networks with  $K$  component dependencies, arrival variance causes expected non-kit wait time  $\mathbb{E}[t_{\text{wait}}]$  and WIP inventory  $I_{\text{WIP}}$  to inflate exponentially:*

$$\mathbb{E}[t_{\text{wait}}] = \int_0^\infty \left( 1 - \prod_{k=1}^K F_k(t) \right) dt \quad (4)$$

*Proof.* Order release requires simultaneous arrival of all  $K$  items. For independent arrival distributions  $F_k(t)$ , the maximum arrival time cumulative distribution function is  $F_{\text{max}}(t) = \prod_{k=1}^K F_k(t)$ . When  $K \gg 1$ , even high single-item arrival probabilities ( $P_k = 0.99$ ) collapse global kitting probability ( $0.99^{100} \approx 0.366$ ), causing  $\mathbb{E}[t_{\text{wait}}] \gg 0$  and inflating WIP via Little's Law ( $I_{\text{WIP}} = \lambda \cdot LT_{\text{shop}}$ ).  $\square$

### 2.5 Theorem 4: Absolute Optimality of Physical-Constraint Truncated IF-THEN-ELSE Stepping Search

**Theorem 2.4.** *For non-convex physical state spaces under Wolfram computational irreducibility, stepping search with inline IF-THEN-ELSE algebraic constraint truncation is the unique strategy ensuring 100% physical feasibility at minimum computational complexity.*

*Proof.* By computational irreducibility, state evolution  $\mathbf{x}_D(t+1) = T_D(\mathbf{x}_D(t), e_t)$  is Turing-complete and cannot be shortcut by analytical closed forms. Unpruned search expands as  $O(B^H)$ . Inline constraint evaluation  $\mathcal{P}_C(S_t) = \mathbb{I}(g(S_t) \leq 0)$  executes instant algebraic truncation (Drop\_Subtree( $S_t$ ) if  $\mathcal{P}_C = 0$ ), pruning non-physical subtrees at depth  $k \ll H$ , collapsing NP-hard complexity to polynomial  $O(N \log N)$  while guaranteeing 100% feasible solutions.  $\square$

## 2.6 Theorem 5: Governance as a Precondition for Observability

**Theorem 2.5.** *In Non-IID entangled networks, rigid algebraic allocation constraints (Governance) must precede observation to ensure mathematical observability and parameter identifiability.*

*Proof.* Unconstrained departmental gaming introduces multi-collinear state dependencies  $x_j(t) = f(x_k(t)) + \epsilon$ , driving state covariance singular ( $\det(\Sigma_{xx}) \rightarrow 0$ ) and estimation variance  $\text{Var}(\hat{\theta}) \rightarrow \infty$ . Imposing rigid allotment constraints  $\Pi \mathbf{x} = \mathbf{b}_t$  forces covariance full rank ( $\det(\Sigma_{xx}) > 0$ ), restoring the Kalman Observability Gramian matrix  $\mathbf{W}_o > 0$  and guaranteeing causal identifiability.  $\square$

## 3 One Plan Architecture, Dimensional Algebra, and DOD Compute Engine

### 3.1 Planning BOM Dimension Algebra

To conquer combinatorial explosion in customized assembly (ATO/CTO), the IPC engine replaces static SKU explosion with seven dimensional attribute operators:

1. **Inheritance ('PASS')**:  $\mathbf{a}_i[\text{key}] \leftarrow \text{PASS}(\mathbf{a}_{\text{parent}}[\text{key}])$ .
2. **Equal Subspace ('EQ')**:  $\mathcal{X}_{\text{alloc}} = \{\mathbf{x}_s \mid \mathbf{x}_s[\text{attr}] = \mathbf{x}_d[\text{attr}]\}$ .
3. **Grade Substitution ('GE'/'LE')**:  $\mathcal{X}_{\text{alloc}} = \{\mathbf{x}_s \mid \mathbf{x}_s[\text{grade}] \geq \mathbf{x}_d[\text{grade}]\}$ .
4. **Range Truncation ('LT'/'GT'/'NE')**: Enforcing trade embargoes and machine capability boundaries.

### 3.2 Allocation-Allotment Dual-Loop Feedback Control

- **Allocation (Slow Loop)**: Mid/long-term capacity pre-reservation mapping demand probabilities to upper-bound quotas:  $\sum_j u_j \leq A_i(t)$ .
- **Allotment (Fast Loop)**: Near-term execution window locking allotments via Demand-by-Demand (DBD) deduction:  $\sum_{j \in O_{\text{firm}}} d_{ji} y_j(t) \leq B_i(t) \leq A_i(t)$ . Administrative rush

orders exceeding available allotment are physically blocked, eliminating secondary disruptions.

### 3.3 Bare-Metal Data-Oriented Design (DOD)

The IPC solver abandons pointer-heavy OOP object graphs:

1. **1D Contiguous Memory Layout:** BOM trees and supply graphs are flattened into 1D continuous arrays, driving L1/L2 cache hit rates to near 100%.
2. **Bitset SIMD Vectorization:** Complex logic decisions are mapped to binary bitsets, leveraging SIMD registers for branch-free parallel execution.

## 4 100-Billion Industrial Empirical Benchmark: LCFC Lighthouse Factory

### 4.1 Experimental Setup

The IPC engine was deployed against full-scale industrial data from LCFC (World Economic Forum Lighthouse Factory):

- **Data Scale:** 500,000 daily discrete demand orders, 2,000,000 SKU-location nodes, 20-level max BOM depth, and 150,000 capacity/machine constraints.
- **Hardware Setup:** Executed on a single commodity personal computer (PC workstation: 1 CPU, 16 cores 3.2GHz, 128GB RAM; solver memory footprint < 12GB).

### 4.2 Empirical Benchmark Results

### 4.3 Verified Financial & Operational Results

Deploying the IPC engine closed-loop resulted in:

- **Delivery Response Rate:** Increased from **54%** to **98%** (stabilized).
- **Order Delivery Accuracy:** Improved to **95% not late, 80% not early**.
- **Inventory & Liquidity:** Reduced structural parts inventory by **50%**, increased overall inventory turnover by  $1.9\times$  (YoY), releasing billions of RMB in working capital.

Table 1: Empirical Performance Comparison: Legacy Commercial Solvers vs. IPC Engine

| Performance Metric                                | Legacy Solvers (ERP/APS) <sup>1</sup>  | Commercial Solvers (Memory Exhaustion) or > 6 Hours | IPC Engine (DOD Bare-Metal)                       |
|---------------------------------------------------|----------------------------------------|-----------------------------------------------------|---------------------------------------------------|
| 500k Orders, 20-Level BOM, 2M+ Nodes Full Netting | System Crash                           | (Memory Exhaustion) or > 6 Hours                    | <b>296 Seconds (5 Minutes) Global Convergence</b> |
| 50k Order Core MRP Netting                        | 3 – 8 Hours                            |                                                     | <b>100.85 Milliseconds</b>                        |
| Memory Architecture                               | Table-driven, OOP Pointer Graphs       |                                                     | Data-Oriented Design (DOD), 1D Array, SIMD        |
| Cache Hit                                         | High Cache Miss Rate (> 85% CPU Stall) |                                                     | <b>100% Cache Hit Rate (Branch-Free)</b>          |
| Auditability                                      | Black-box statistical aggregates       |                                                     | <b>100% Causal Traceability (DuckDB Sandbox)</b>  |

<sup>1</sup> Legacy commercial solvers refer to leading enterprise APS/ERP engines (e.g., SAP APO/IBP, Oracle ASCP, Blue Yonder).

## 5 Discussion: Unification with Five Classical System Theories

Value Chain Physics provides a root evolution equation:

$$V_{\Omega}(t+1) = M \cdot \Pi \left[ \sum_{i=1}^n (m_i^* \cdot \pi_i^* \cdot S_i) + \Delta(t) \right] \quad (5)$$

This equation unifies five classic systems paradigms:

1. **Prigogine Dissipative Structures:** Idempotent projection  $\Pi$  ( $\Pi^2 = \Pi$ ) acts as a negative entropy pump ( $\frac{d_e S}{dt} = \text{tr}(\nabla_{\mathbf{S}}(\Pi \mathbf{S})) < 0$ ), neutralizing internal entropic dissipation.
2. **Ashby Requisite Variety:** When controller variety  $V_c = \dim(\text{Im}(\Pi)) < V_s \approx O(N!)$ , residual accumulates as  $\lim_{t \rightarrow \infty} \|\Delta(t)\| = \infty$ . Balance is restored when  $\Pi$  covers the state support.
3. **Friston Free Energy Principle:** Residual norm  $\Delta(t)$  represents surprise. State updates execute belief update; action selection minimizes expected future surprise  $\mathbf{a}^* = \arg \min_{\mathbf{a}} \mathbb{E}_q[\|\Delta(t+1)\|^2]$ .
4. **Conant-Ashby Good Regulator Theorem:** Model divergence  $\epsilon_{\text{model}}$  imposes a lower bound on residual  $\lim_{t \rightarrow \infty} \|\Delta(t)\| \geq \frac{\dim(\text{Ker}(h)) \|\epsilon_{\text{model}}\|}{1 - \|\Pi\|} > 0$ , proving pure black-box AI without physical models is doomed to collapse.
5. **Renormalization Group Theory:** Operator  $\Pi$  acts as a coarse-graining operator, integrating micro-interactions  $m_i$  into macro effective coupling  $m_i^*$ .

## 6 Conclusion

Value Chain Physics establishes that enterprise efficiency dissipation is not a managerial failure, but a physical inevitability of violating underlying system laws. By replacing open-loop human intervention with silicon-based bare-metal algebraic pruning, VCP provides a computable, verifiable, and falsifiable hard physics constitution for open complex giant systems.

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## Appendix: Formal Pseudocode Algorithms

### Appendix A: Branch-Free Priority Netting

Algorithm 1: Branch-Free Priority Netting

-----  
Input: Orders set  $O = \{o_1, o_2, \dots, o_m\}$ , Supply time-series  $S = [s_1, \dots, s_T]$   
Output: Updated Supply  $S$ , and Allocated quantities.

1. For each order  $o_i$  in  $O$ :
  2.     Pack multi-dimensional priorities into 64-bit unsigned integer:  
        $P_i \leftarrow (\text{committed\_bit} \ll 62) \mid (\text{tier\_bits} \ll 60) \mid (\text{due\_bits} \ll 44) \mid$   
           $(\text{pri\_bits} \ll 28) \mid (M_{\text{rev}} - \min(M_{\text{rev}}, \text{Revenue}_i))$
  3. Sort  $O$  in ascending order of  $P_i$  using SIMD radix sort.
  4. For each sorted order  $o_i$  in  $O_{\text{sorted}}$ :
  5.      $D_{\text{req}} \leftarrow o_i.\text{quantity}$ ;  $t_{\text{due}} \leftarrow o_i.\text{due\_day}$
  6.     For  $t = t_{\text{due}}$  down to 0:
  7.          $q_{\text{alloc}} \leftarrow \text{std::min}(S[t], D_{\text{req}})$      // Branch-free CMOV
  8.          $S[t] \leftarrow S[t] - q_{\text{alloc}}$
  9.          $D_{\text{req}} \leftarrow D_{\text{req}} - q_{\text{alloc}}$
  10.         $o_i.\text{allocated}[t] \leftarrow q_{\text{alloc}}$
  11.        If  $D_{\text{req}} \leq 0$  Then Break
- 

### Appendix B: SIMD Parallel Prefix Scan & Segment Tree Merger

Algorithm 2: Parallel Prefix Scan \& Segment Tree Associative Merger

-----  
Input: Daily net supply/demand array  $\text{Net}[T]$ , Topology matrix  $\text{eta}[K][K]$   
Output: OnHand time-series and global deficit/surplus vectors.

1. Parallel Blelloch Scan over time horizon  $T$  (Time:  $O(\log T)$ )
2. Associative Merger Operator (oplus) for Segment Tree Nodes:  
    $u_{\text{Parent}} = u_L \text{ opus } u_R$
3. Match deficits of  $R$  with surpluses of  $L$  along DAG paths:  
   For site  $i$  in  $[0, K-1]$ :  
      For site  $j$  in  $\text{Parents}(i)$ :  
           $q_{\text{trans}} \leftarrow \min(\text{Deficit}_R[i], \text{Surplus}_L[j] * \text{eta}[j][i])$   
           $\text{Deficit}_R[i] \leftarrow \text{Deficit}_R[i] - q_{\text{trans}}$   
           $\text{Surplus}_L[j] \leftarrow \text{Surplus}_L[j] - q_{\text{trans}} / \text{eta}[j][i]$
4. Return  $u_{\text{Parent}} = (\text{Deficit}_{\text{Parent}}, \text{Surplus}_{\text{Parent}})$

---

## Appendix C: OTP Bidirectional Scheduling & Zero-Heap Rollback Stack

Algorithm 3: OTP Scheduling \& Thread-Local Zero-Heap Stack

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Input: Order (due\_day, duration, wc\_id), CapGrid[WC\_Num \* Horizon], TLS\_Stack

Output: Planned delivery day or rejection.

```
1. PSD <- -1
2. For t = request_day to Horizon - 1:
    If CapGrid[wc_id * Horizon + t] >= duration Then:
        PSD <- t; Break
3. If PSD == -1 Then Return "Insufficient Capacity"
4. TLS_Stack.Clear()
5. CapGrid[wc_id * Horizon + PSD] -= duration
6. TLS_Stack.Push(wc_id, PSD, duration)
7. TLS_Stack.Commit() // O(1) clear, accept schedule
8. Return PSD
```

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